

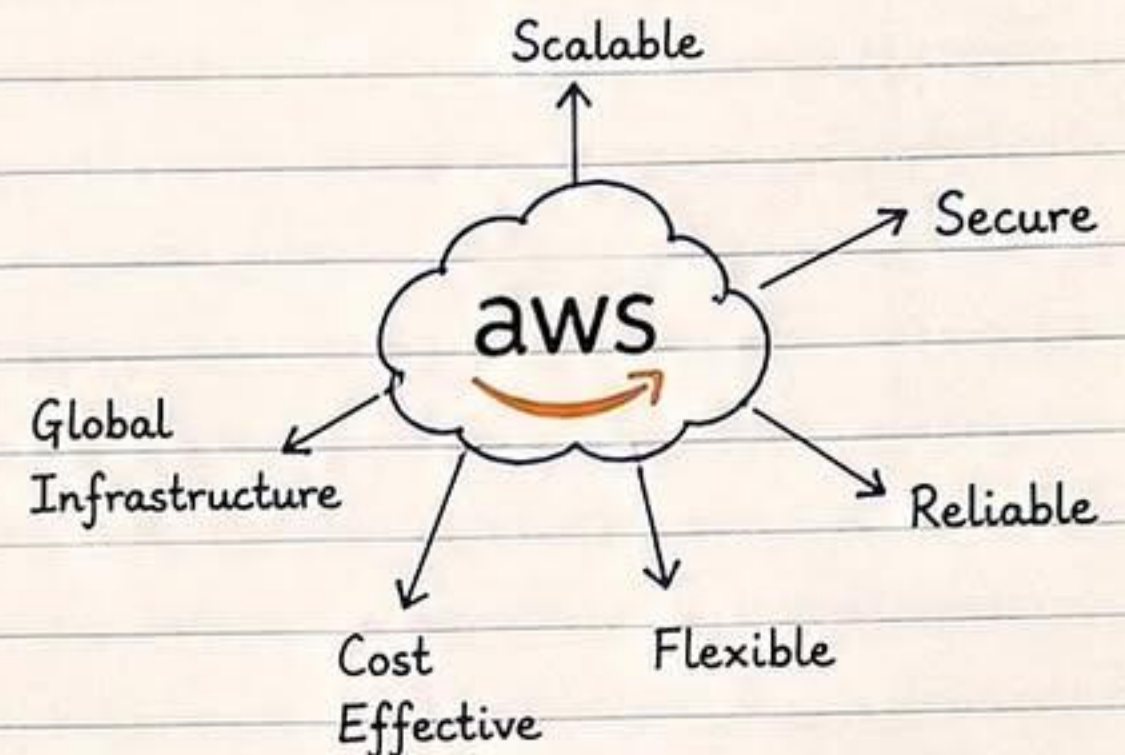
AWS Notes



Amazon Web Services - Cloud Computing Platform

* What is AWS?

AWS (Amazon Web Services) is a cloud computing platform provided by Amazon. It offers a wide range of services like compute, storage, database, networking, security, machine learning, analytics etc. on demand over the internet.

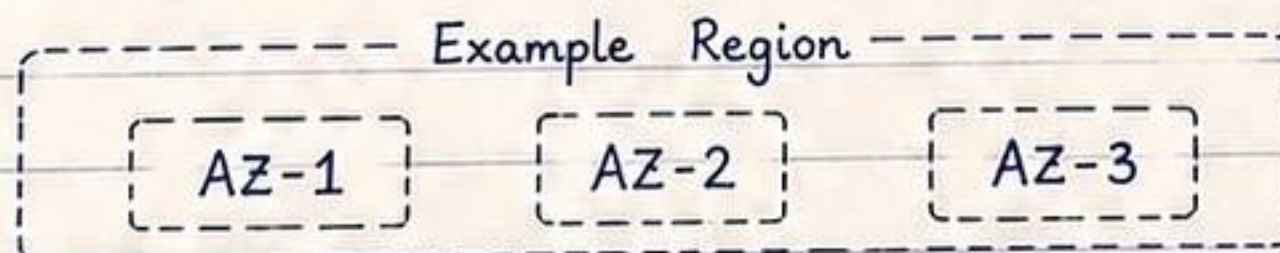


* Key Benefits of AWS

- No upfront cost
- Pay-as-you-go
- Highly scalable
- High availability & reliability
- Global infrastructure
- Security and compliance

* AWS Global Infrastructure.

AWS has data centers in many Regions around the world. Each Region has multiple Availability Zones (AZs) which are isolated data centers.



* AWS Service Categories

Category	Description
Compute	Virtual servers and containers to run applications.
Storage	Scalable storage services for all types of data.
Database	Managed databases (SQL, NoSQL, etc.).
Networking	Networking services like VPC, Route 53, CloudFront.
Security	Identity management, encryption, threat protection.
Analytics	Big data processing, data warehousing, ML services.
Management	Monitoring, automation, logging, cost management.

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AWS Global Infrastructure

AWS has a global infrastructure designed to run your applications with high availability, fault tolerance and low latency.

1. Region

A Region is a physical location in the world where AWS has multiple data centers.

- Example : us-east-1 (N. Virginia)

2. Availability Zone (AZ)

An AZ is one or more discrete data centers with redundant power, networking and connectivity in a Region.

- A Region has multiple AZs (usually 3 or more).
- AZs are isolated from each other.

3. Edge Location

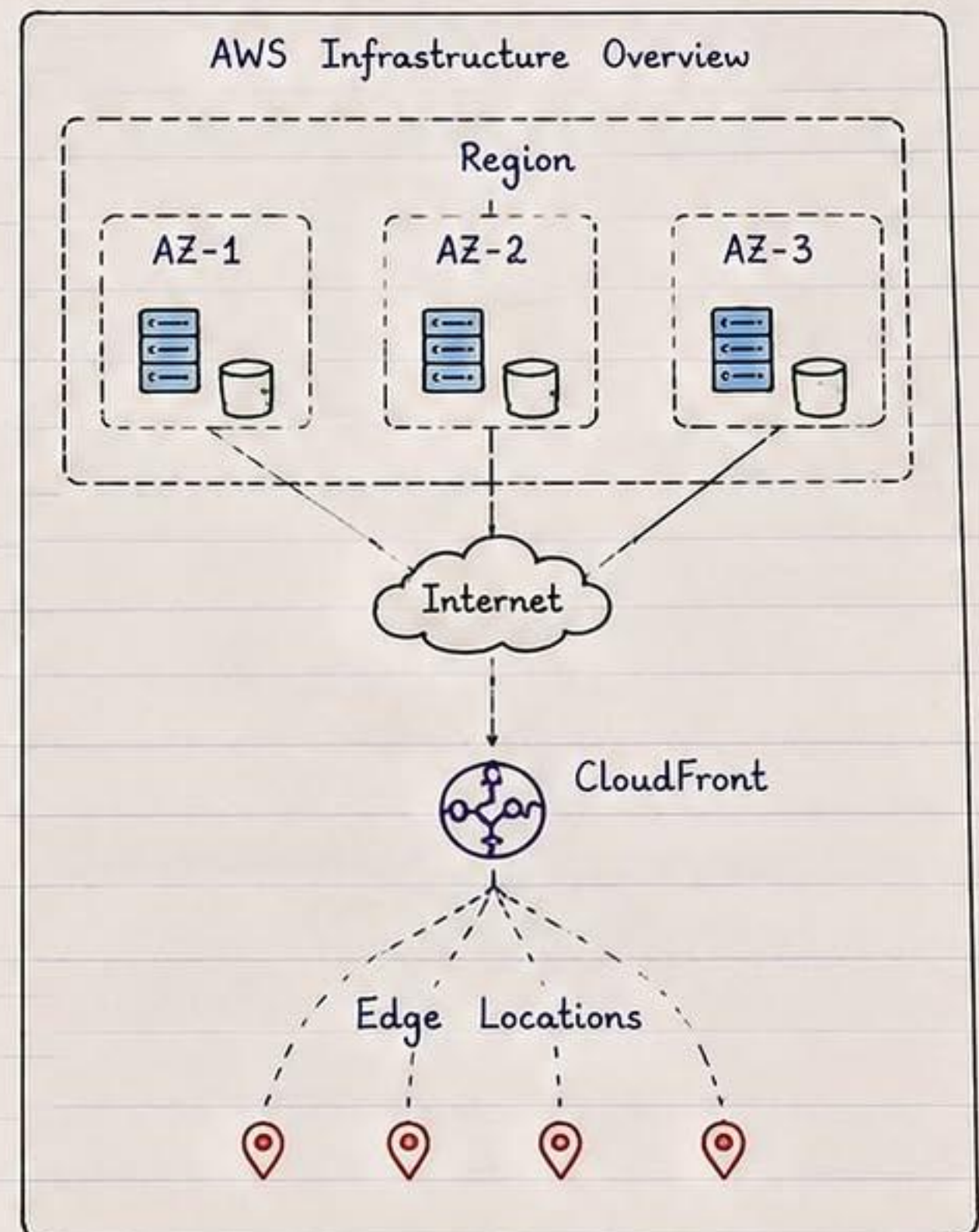
Edge Locations are sites used by CloudFront to cache content closer to users for low latency delivery.

4. Local Zones

Local Zones are extensions of a Region that place compute, storage, database and other services closer to large population centers.

5. Wavelength Zones

Wavelength Zones are AWS infrastructure deployments that bring AWS services even closer to 5G mobile users.



Why this matters ?

- Improves High Availability
- Supports Fault Tolerance
- Enables Disaster Recovery
- Reduces Latency for users worldwide

Production Tip



Always design your applications to span multiple AZs in a Region to avoid a single point of failure.

Interview Question ?

Q. What is the difference between Region and Availability Zone ?

Ans: Region is a physical location in the world, while AZ is a isolated data center

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Core Compute Services

AWS provides a wide range of compute services to run your applications based on different requirements.

1. Amazon EC2 (Elastic Compute Cloud)

- Resizable compute capacity in the cloud.
- Full control over operating system, storage, networking, and security.
- Supports Windows, Linux, and more.

2. AWS Lambda

- Serverless compute service.
- Run code without provisioning or managing servers.
- Pay only for the time your code runs.

3. Amazon ECS (Elastic Container Service)

- Fully managed container orchestration service.
- Supports Docker containers.
- Integrates with other AWS services.

5. Elastic Beanstalk

- Easy way to deploy and manage applications.
- Supports multiple languages and platforms.
- Automatically handles capacity provisioning, load balancing, scaling, and health monitoring.

7. Elastic Load Balancer (ELB)

- Distributes incoming traffic across multiple targets.
- Improves availability and fault tolerance.
- Types: ALB (Layer 7), NLB (Layer 4), Gateway Load Balancer.

Production Tip 💡

- Use Auto Scaling with ELB to handle traffic spikes.
- Prefer managed services (ECS, EKS, Lambda) to reduce operational overhead.

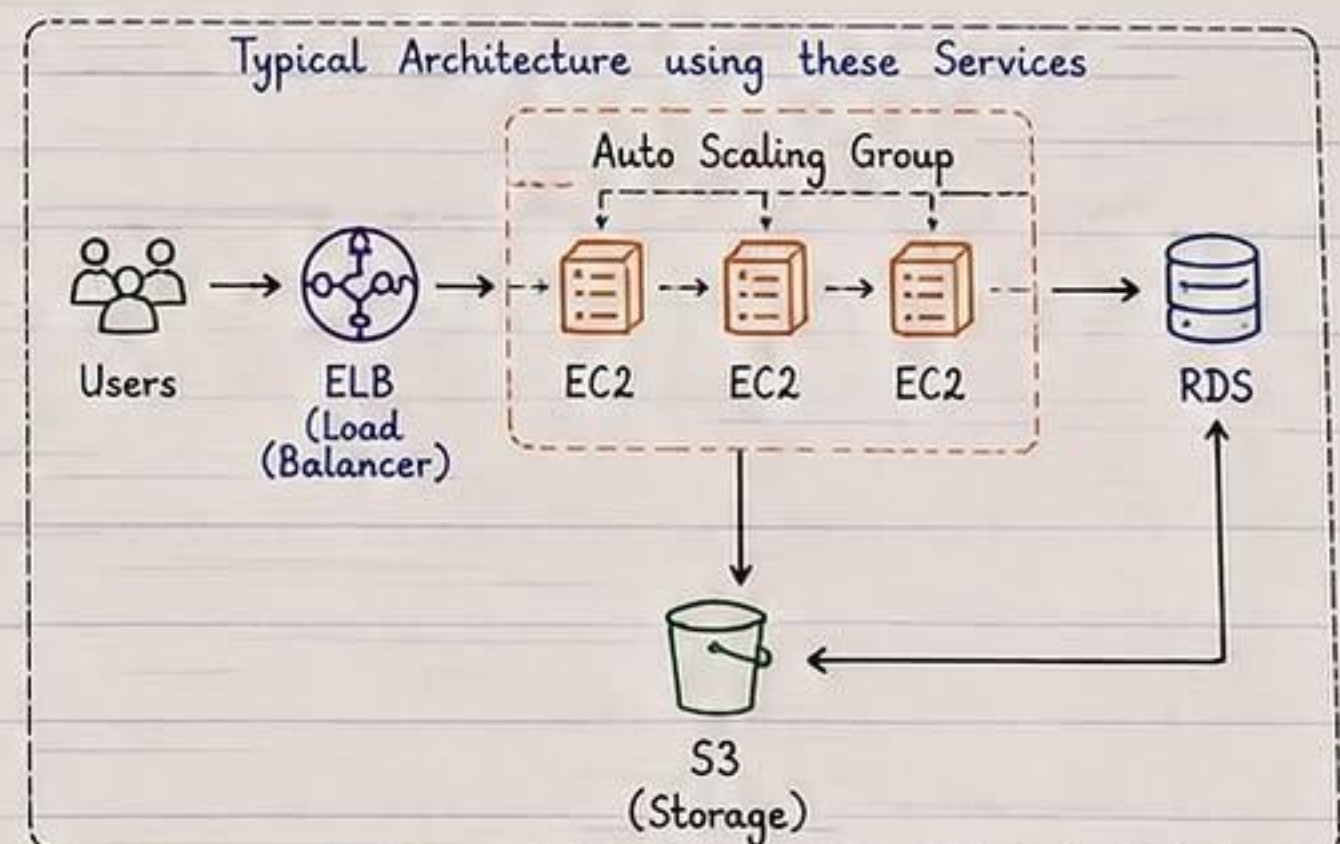
Core Compute Services Overview	
Service	Use Case
EC2	Virtual Machines in the cloud
Lambda	Run code without managing servers
ECS	Run and manage Docker containers
EKS	Run Kubernetes clusters
Elastic Beanstalk	Easy deployment of applications
Auto Scaling	Automatically adjust capacity
Elastic Load Balancer	Distribute traffic across targets

4. Amazon EKS (Elastic Kubernetes Service)

- Managed Kubernetes service.
- Run Kubernetes without installing and operating it.
- Highly available and scalable.

6. Auto Scaling

- Automatically adjusts the number of instances based on demand.
- Helps maintain application availability and optimize cost.



Interview Question ?

Q. When would you use EC2 over Lambda?

Ans: Use EC2 for long-running applications, full OS control, custom configurations.

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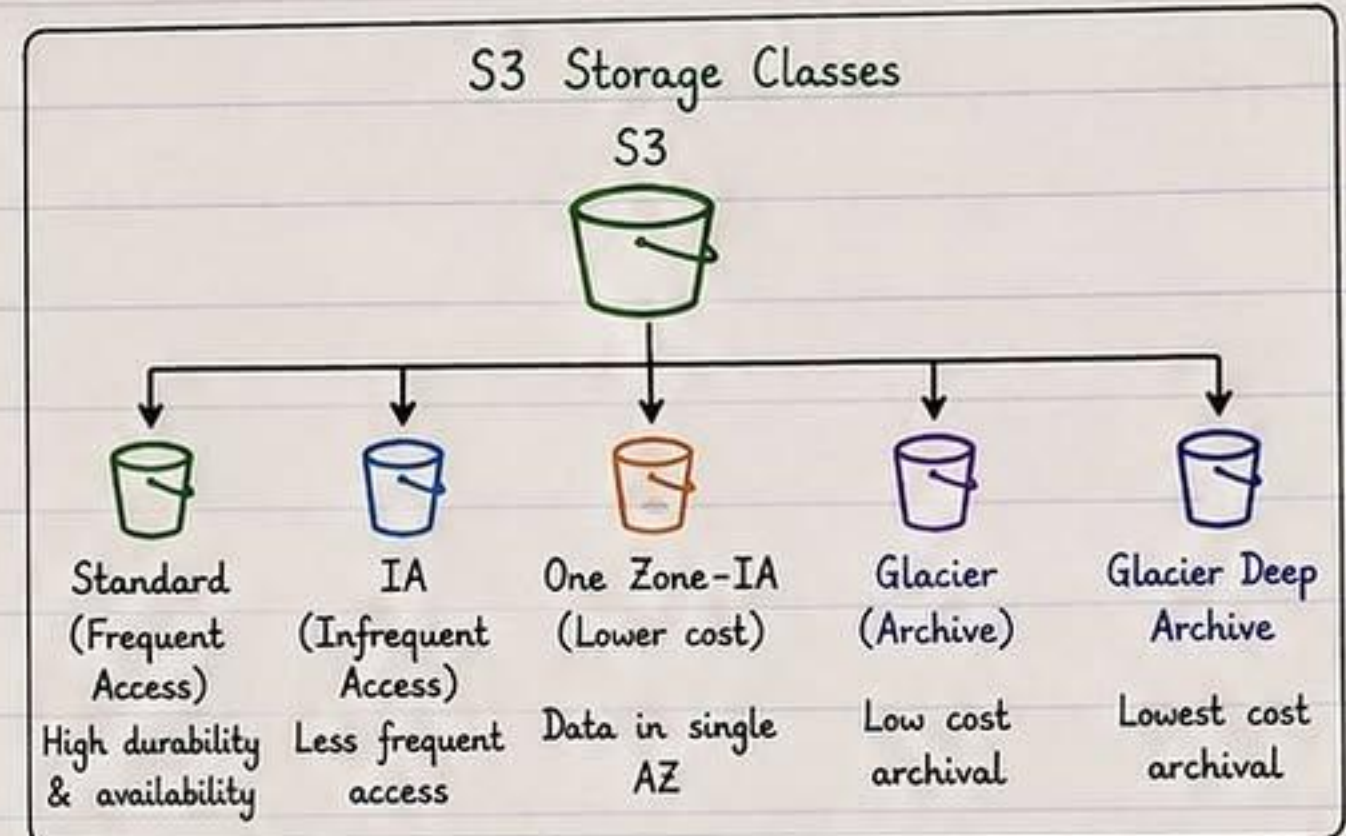


Storage Services

AWS provides scalable, durable and secure storage services to store any amount of data for any use case.

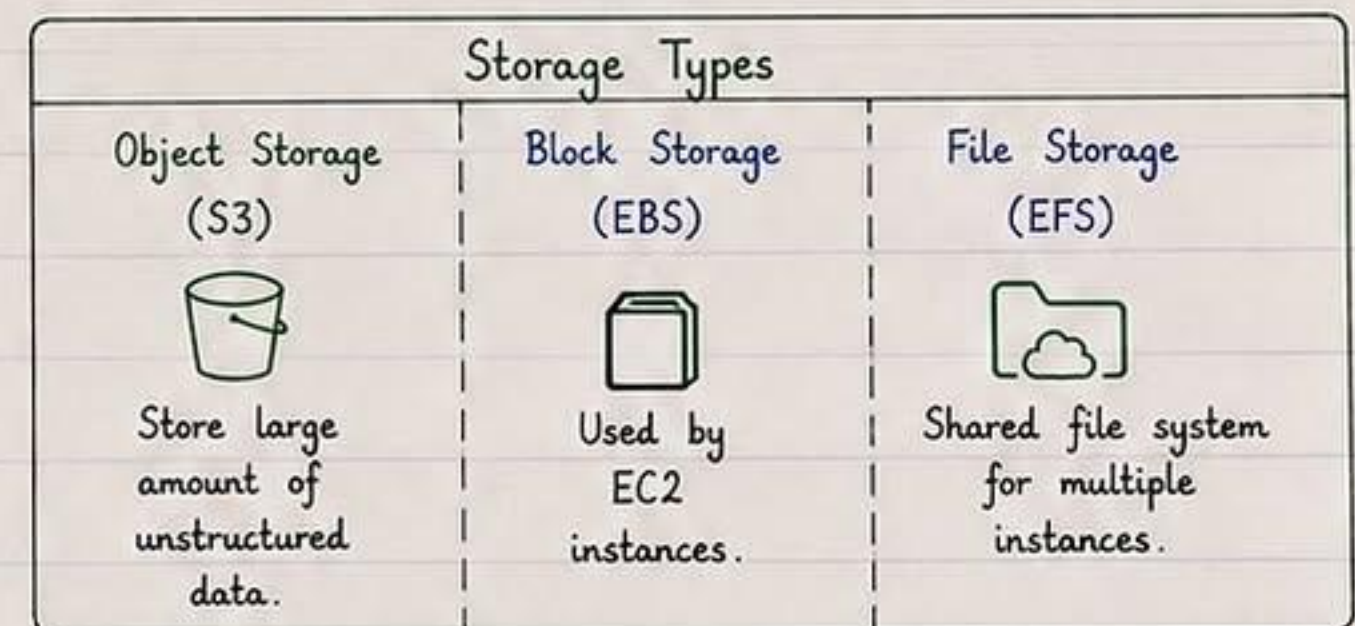
1. Amazon S3 (Simple Storage Service)

- Object storage service to store and retrieve any amount of data from anywhere.
- Highly durable, available and scalable.
- Common use cases : backups, static website hosting, data lakes, media storage.



2. Amazon EBS (Elastic Block Store)

- Block storage for EC2 instances.
- Data persists even if the instance stops.
- Used for OS, databases and applications.
- Types : gp3 (general purpose), io2 (high performance), st1 (throughput optimized).



3. Amazon EFS (Elastic File System)

- Managed file storage for use with AWS cloud services and on-premises.
- Multiple EC2 instances can access the file system simultaneously.
- Use cases : Content management, Web serving, Home directories.

4. Amazon Glacier

- Long-term archival storage.
- Very low cost.
- Retrieval time : Minutes to Hours.
- Use for data that is rarely accessed.

5. Storage Gateway

- Hybrid storage service to connect on-premises environments with AWS cloud storage.
- Types : File Gateway, Volume Gateway, Tape Gateway.

Production Tip

- Use S3 Lifecycle policies to move data to IA / Glacier to optimize cost.
- Choose the right storage type based on access pattern and performance needs.

6. AWS Storage Comparison

Service	Type	Use Case	Access	Durability
S3	Object	Backups, Media, Data Lake	HTTP/HTTPS	11 9's
EBS	Block	EC2, OS, DB	Attached to EC2	9 9's
EFS	File	Shared Files	NFS	11 9's
Glacier	Object (Archive)	Long-term Archive	Retrieval	11 9's

Interview Question

Q. When should you use EBS vs S3 ?

Ans: Use EBS for block-level storage attached to EC2 instances (OS, DB).

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Database Services

AWS offers fully managed database services for relational, non-relational, in-memory caching, data warehouse and more.

1. Amazon RDS (Relational Database Service)

- Managed relational databases in the cloud.
- Supports engines: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora.
- Features: Automated backups, Multi-AZ, Read Replicas, Easy scaling.

2. Amazon Aurora

- AWS designed for the cloud, MySQL and PostgreSQL compatible.
- High performance, availability and durability.
- Up to 5x performance of MySQL on RDS.

3. Amazon DynamoDB

- Fully managed NoSQL key-value and document database.
- Single-digit millisecond performance at any scale.
- Built-in security, backups and global tables.

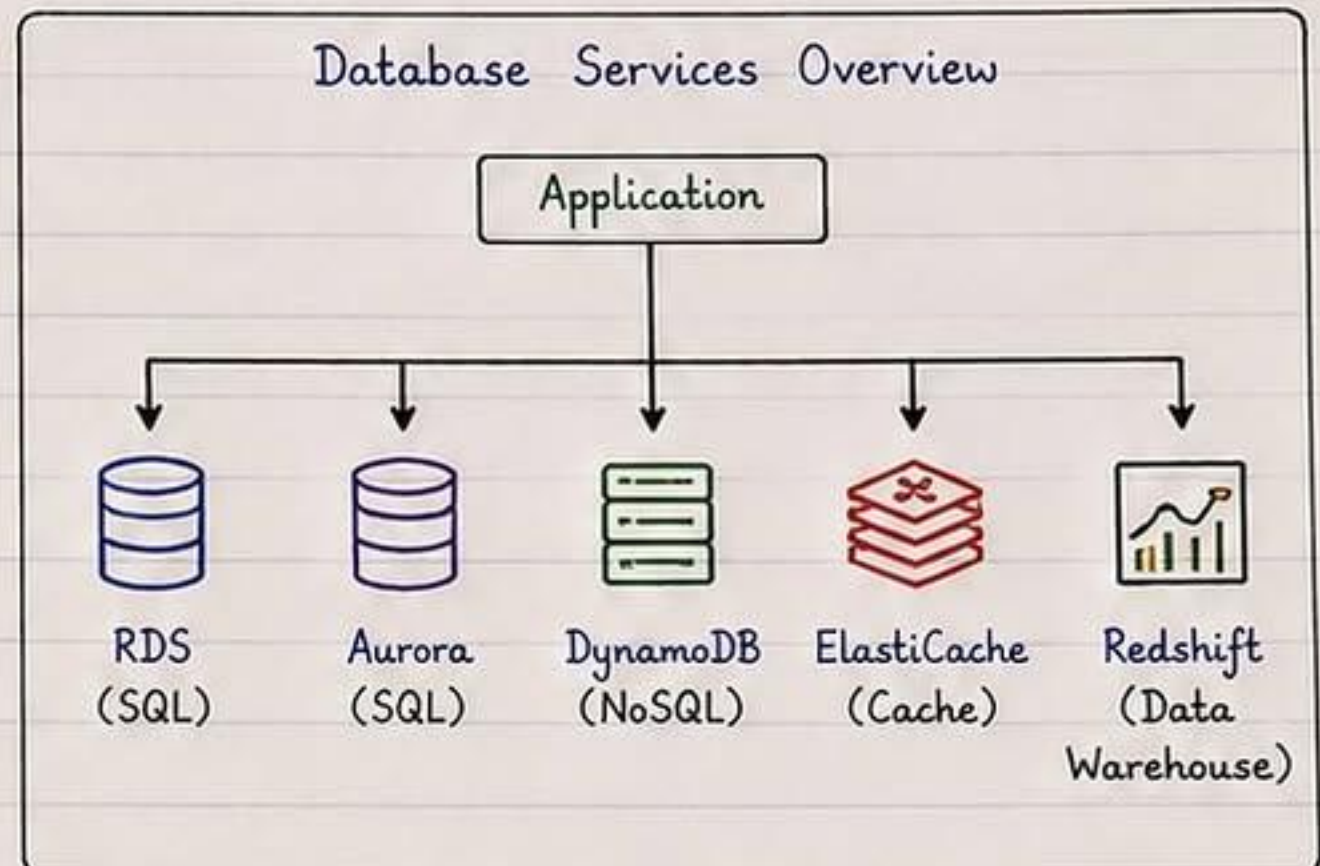
4. Amazon ElastiCache

- Managed in-memory caching service.
- Supports Redis and Memcached.
- Improve application performance and reduce load on databases.

5. Amazon Redshift

- Fully managed data warehouse service.
- Columnar storage for analytics workloads.
- Integrates easily with data lakes, BI tools.

Database Services Overview



SQL vs NoSQL (DynamoDB)

Feature	RDS / Aurora (SQL)	DynamoDB (NoSQL)
Data Model	Tables (Rows & Columns)	Key-Value / Document
Schema	Fixed Schema	Schema-less
Scaling	Vertical & Read Replicas	Automatic & Horizontal
Query Language	SQL	NoSQL API / PartiQL
Transactions	ACID	Eventually Consistent (Optional Strong)
Best For	Complex queries, relationships	High scale, low latency, flexible data
Example Use Cases	E-commerce, CMS, Finance Apps	Gaming, IoT, Sessions, Cart Data

Production Tips 💡

- Use RDS/Aurora for transactional workloads.
- Use DynamoDB for high scale, low latency and unstructured data.
- Enable Multi-AZ for high availability.
- Use ElastiCache to cache frequently accessed data.
- Regularly enable automated backups and monitoring.

Common Use Cases

- RDS** : User management, billing, product catalog
- Aurora** : High traffic web applications
- DynamoDB** : Shopping cart, leaderboards, IoT data
- ElastiCache** : Session store, caching DB queries
- Redshift** : Business analytics, reporting, dashboards

Interview Question ❓

Q. When would you use DynamoDB instead of RDS?

Ans: Use DynamoDB when you need massive scalability, single-digit millisecond latency

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Networking in AWS

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AWS provides secure, scalable and highly available networking services to build your cloud infrastructure.

1. Amazon VPC (Virtual Private Cloud)

- A logically isolated section of AWS cloud where you can launch AWS resources.
- You have full control over IP address range, subnets, route tables and gateways.

2. Subnets

- Subnets are segments within a VPC.
- Types :
 - Public Subnet : Has a route to Internet Gateway.
 - Private Subnet : No direct access to Internet.

3. Internet Gateway (IGW)

- Allows communication between your VPC and the internet.
- Required for public resources.

4. NAT Gateway

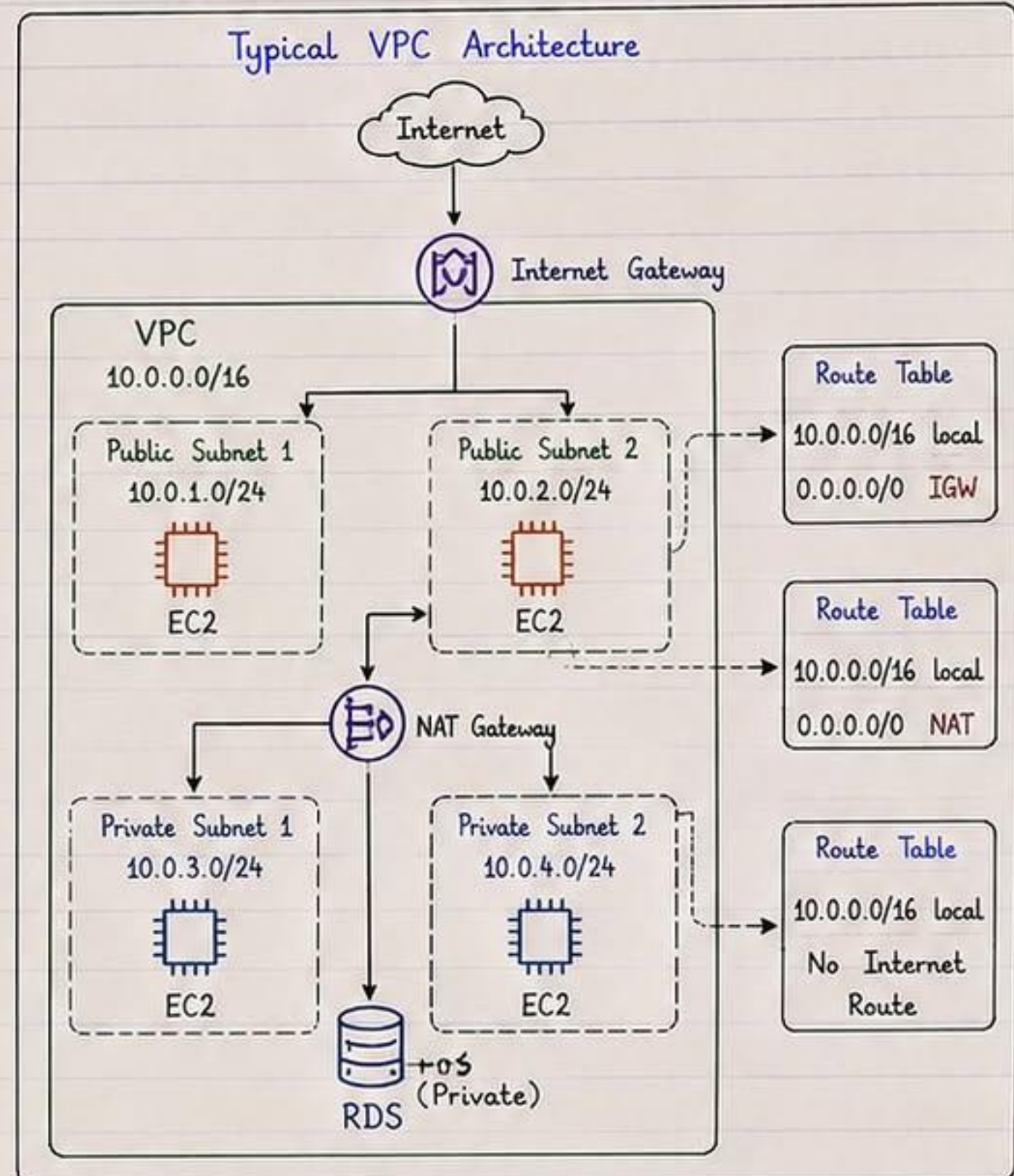
- Allows instances in private subnets to access the internet.
- But blocks incoming traffic from internet.

5. Route Tables

- Contains rules (routes) that determine where network traffic is directed.
- Each subnet must be associated with a route table.

6. Security Groups

- Acts as a virtual firewall at instance level.
- Stateful : Allows return traffic automatically.
- Only allows traffic (Allow rules).



7. Network ACL (NACL)

- Acts as a firewall at subnet level.
- Stateful : Must allow both inbound and outbound traffic.
- Can be associated with multiple subnets.

8. VPC Peering

- Connects two VPCs privately using AWS network.
- Helps in secure communication between VPCs.

Production Tips

- Use private subnets for databases and backend services.
- Keep NAT Gateway in public subnet.
- Use Security Groups instead of NACLs unless required.
- Enable VPC Flow Logs for monitoring traffic.

Quick Comparison

Feature	Security Group	Network ACL
Level	Instance Level	Subnet Level
Stateful	Yes	No
Rules	Allow Only	Allow + Deny
Return Traffic	Automatically allowed	Must be allowed
Default	Allow all outbound	Deny all

Interview Question

Q. What is the difference between Security Group and NACL ?

Ans: Security Group operates at instance level, is stateful and allows traffic.

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Security & IAM

AWS provides a shared responsibility model. Security of the cloud is AWS's responsibility and security in the cloud is your responsibility.

1. IAM (Identity and Access Management)

- IAM helps you securely control access to AWS services and resources.
- You can create Users, Groups, Roles and apply Policies.

2. IAM Users

- Represent people or applications that need long-term access.
- Access using username & password or access keys (Access Key ID & Secret Key).

3. IAM Groups

- A collection of IAM users.
- Permissions are applied to groups, not to individual users.

4. IAM Roles

- Used by AWS services or applications to access AWS resources securely.
- No long-term credentials.

5. IAM Policies

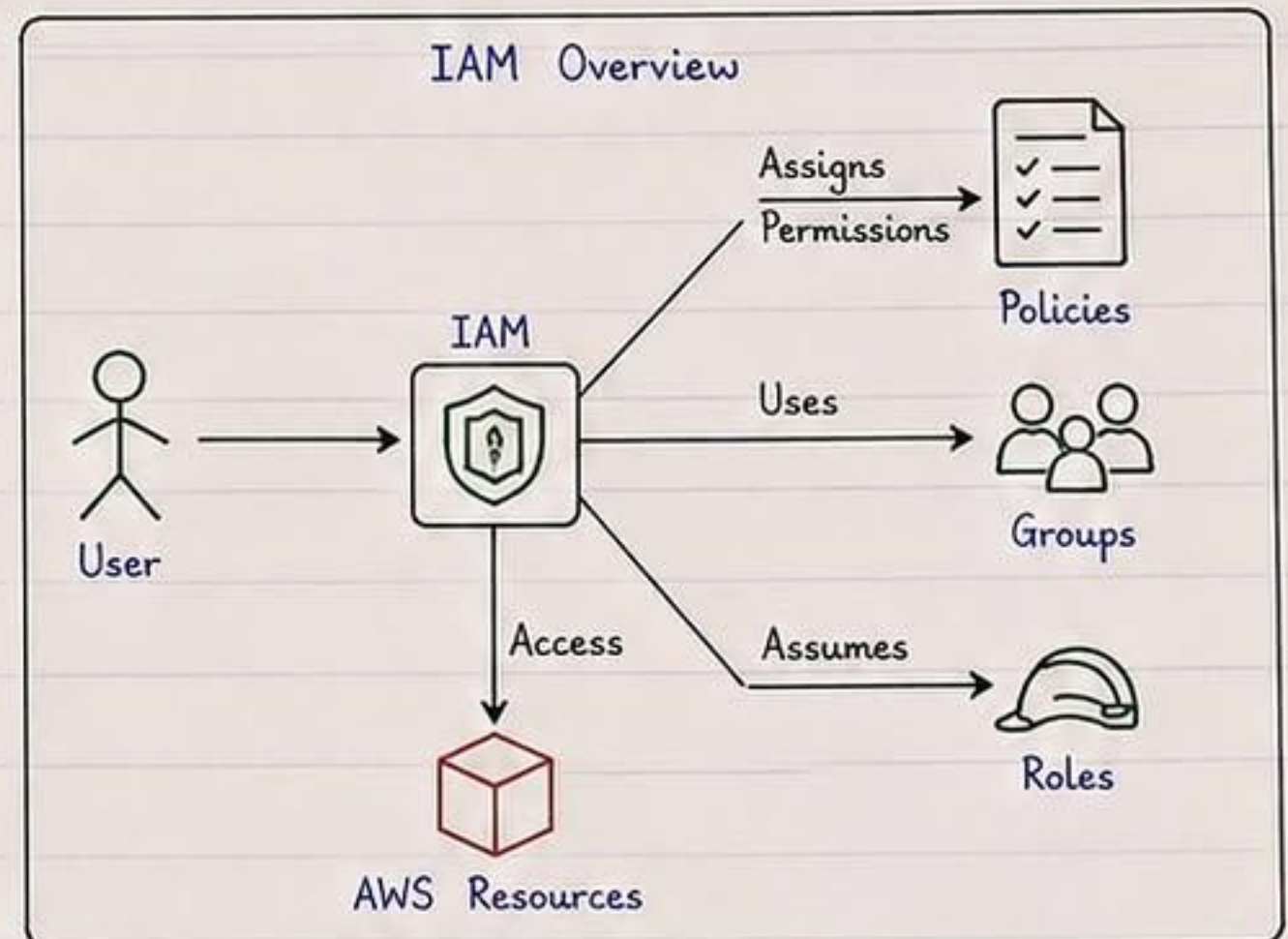
- JSON documents that define permissions.
- Types :
 - Managed Policies (AWS or Customer)
 - Inline Policies
 - Permission Boundaries

6. Multi-Factor Authentication (MFA)

- Adds an extra layer of security.
- Recommended for all root and IAM users.

Production Tips

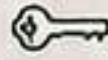
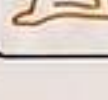
- Never use root user for daily tasks.
- Use least privilege access.
- Enable MFA for all IAM users.
- Rotate access keys regularly.
- Use IAM Roles for EC2, Lambda and other AWS services.
- Enable CloudTrail for full audit logs.



Shared Responsibility Model

AWS (Security of the Cloud)	You (Security in the Cloud)
• Global Infrastructure • Regions, AZs, Edge Locations • Physical Security • Hardware, Networking, Storage • Virtualization Layer • Managed Services (RDS, S3, DynamoDB, etc.)	• Data • IAM (Users, Roles, Policies) • Operating System • Network Configuration • Security Groups, NACLs • Application Code • Monitoring & Logging

Key Security Services

	AWS KMS	: Manage encryption keys for data protection.
	AWS Secrets Manager	: Securely store and manage secrets like API keys, DB passwords.
	AWS Shield	: Protects applications from DDoS attacks.
	AWS CloudTrail	: Logs API calls and user activity.
	AWS Inspector	: Automatically assesses security vulnerabilities.

Interview Question ?

Q. What is the difference between IAM User and IAM Role ?

Ans: IAM User represents a person or application with long-term access.

IAM Role is used by AWS services or applications to get temporary access without

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Monitoring & DevOps

AWS provides powerful monitoring and DevOps services to help you automate, deploy and monitor your applications.

1. Amazon CloudWatch

- Monitor resources, collect metrics and logs.
- Create dashboards and set alarms.
- Monitor EC2, RDS, Lambda, APIs and more.

2. AWS CloudTrail

- Records API calls and user activity.
- Helps with security analysis and compliance.
- Integrates with CloudWatch and S3.

3. Amazon SNS

- Simple Notification Service.
- Sends notifications via Email, SMS, HTTP, Lambda and more.

4. Amazon SQS

- Simple Queue Service.
- Decouples and scales microservices.
- Ensures reliable message delivery.

5. AWS CodePipeline

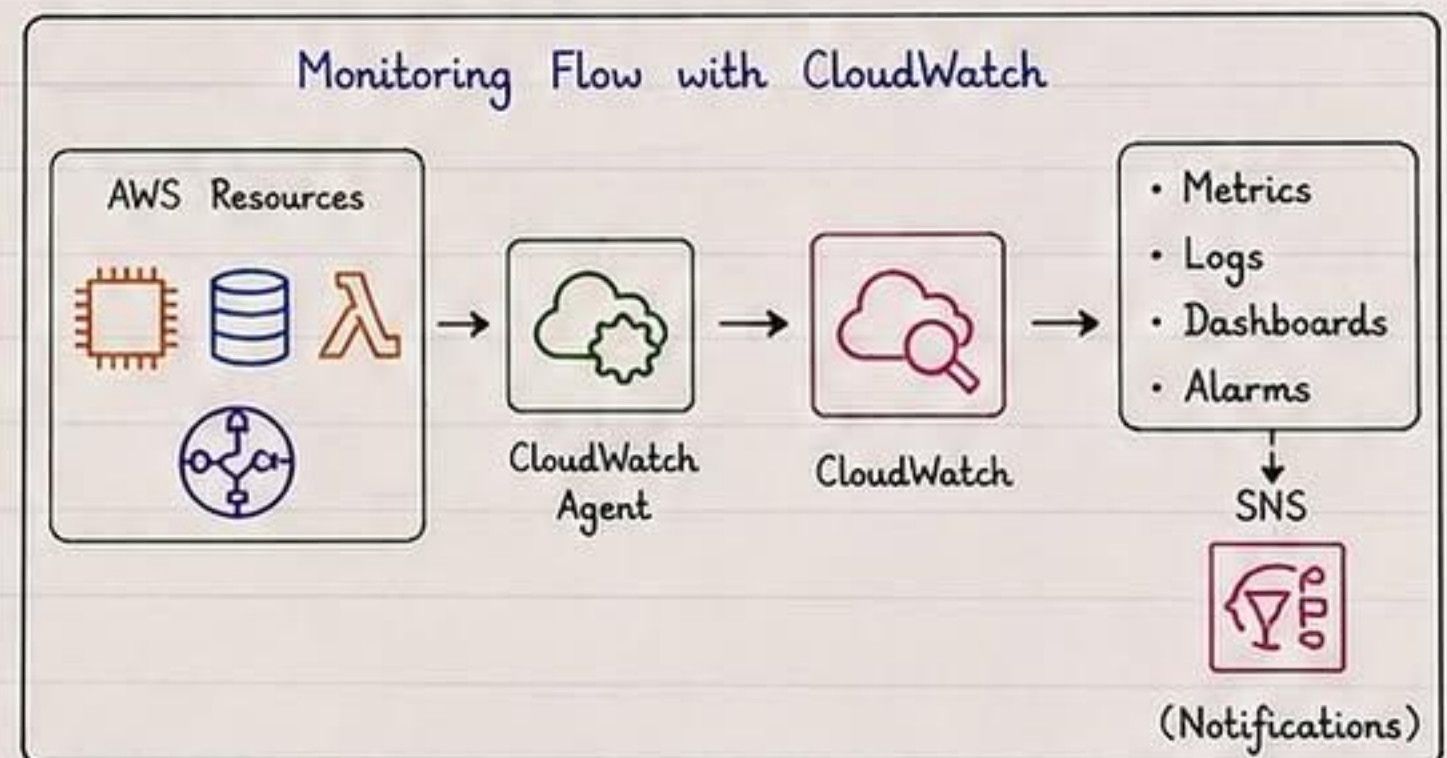
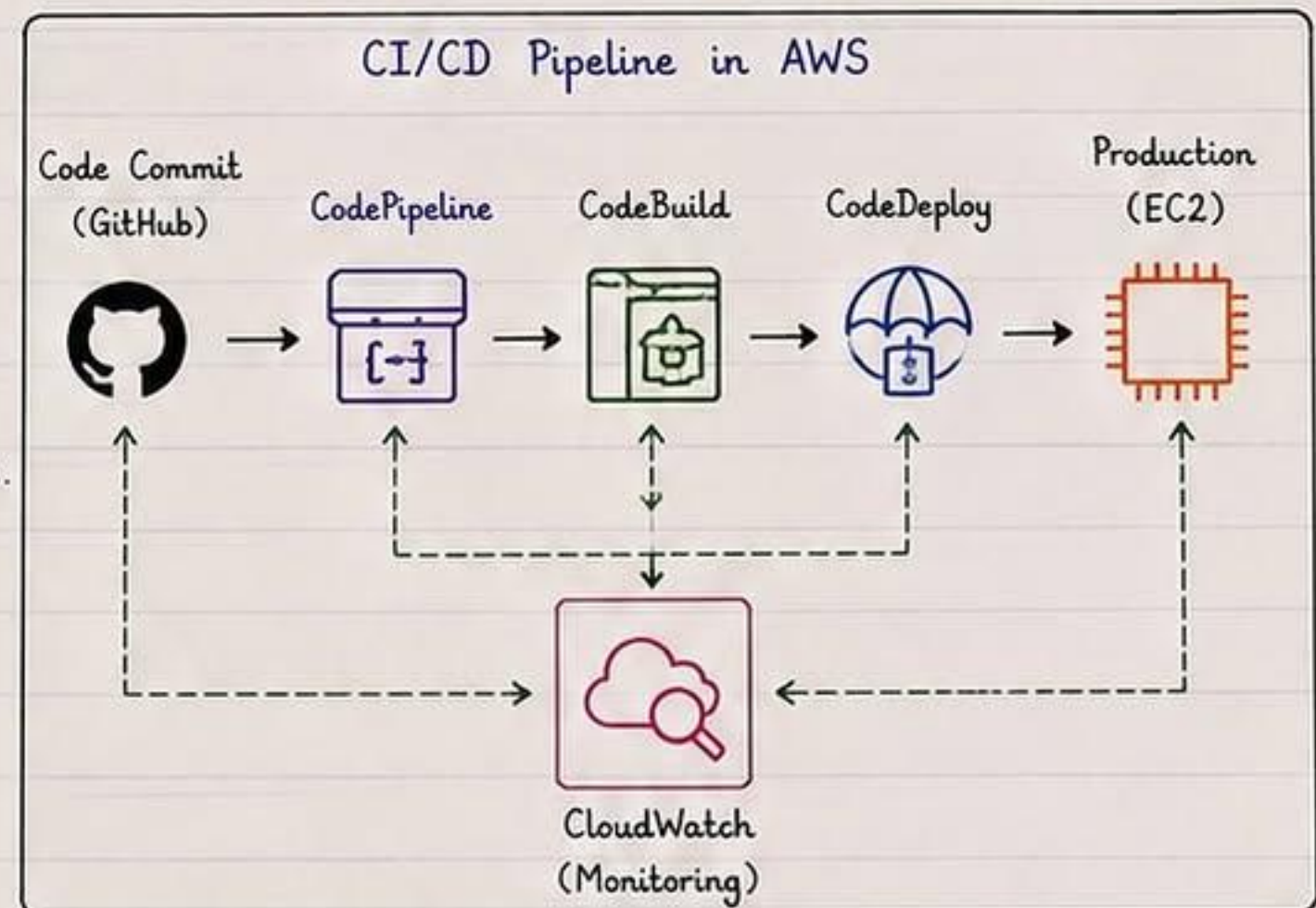
- Automates build, test and deployment process.
- Orchestrates the entire CI/CD pipeline.

6. AWS CodeBuild

- Compiles source code and runs tests.
- Supports multiple languages and build environments.

7. AWS CodeDeploy

- Automates application deployments.
- Supports EC2, Lambda and on-premises.
- Rolling, Blue/Green and Canary deployments.



Key Services Summary

Service	Purpose
CloudWatch	Monitoring, Metrics, Logs, Alarms, Dashboards
CloudTrail	Audit logs of API calls and user activity
SNS	Push notifications to users & systems
SQS	Message queue for decoupling services
CodePipeline	Automate build, test and deployment
CodeBuild	Build and test your code
CodeDeploy	Automate deployments to environments

Production Tips 💡

- Always enable CloudTrail and send logs to S3.
- Set CloudWatch Alarms for critical metrics.
- Use SNS for instant alerts on failures.
- Use SQS to handle traffic spikes and retries.
- Automate everything with CI/CD pipeline.

Interview Question ?

Q. How does CloudWatch help in monitoring an application ?

Ans: CloudWatch collects metrics, logs and events from AWS resources and applications. It allows you to create dashboards, set alarms and get notified via SNS when

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Architecture Patterns & Best Practices

AWS provides proven architecture patterns to build scalable, resilient and highly available applications.

1. Three-Tier Architecture

- Separates application into Presentation, Application and Data layers.
- Improves security, scalability and maintainability.

2. High Availability

- Deploy resources across multiple Availability Zones.
- Use ELB, Auto Scaling and RDS Multi-AZ.

3. Auto Scaling

- Automatically adjusts the number of instances based on demand.
- Improves performance and reduces cost.

4. Load Balancing

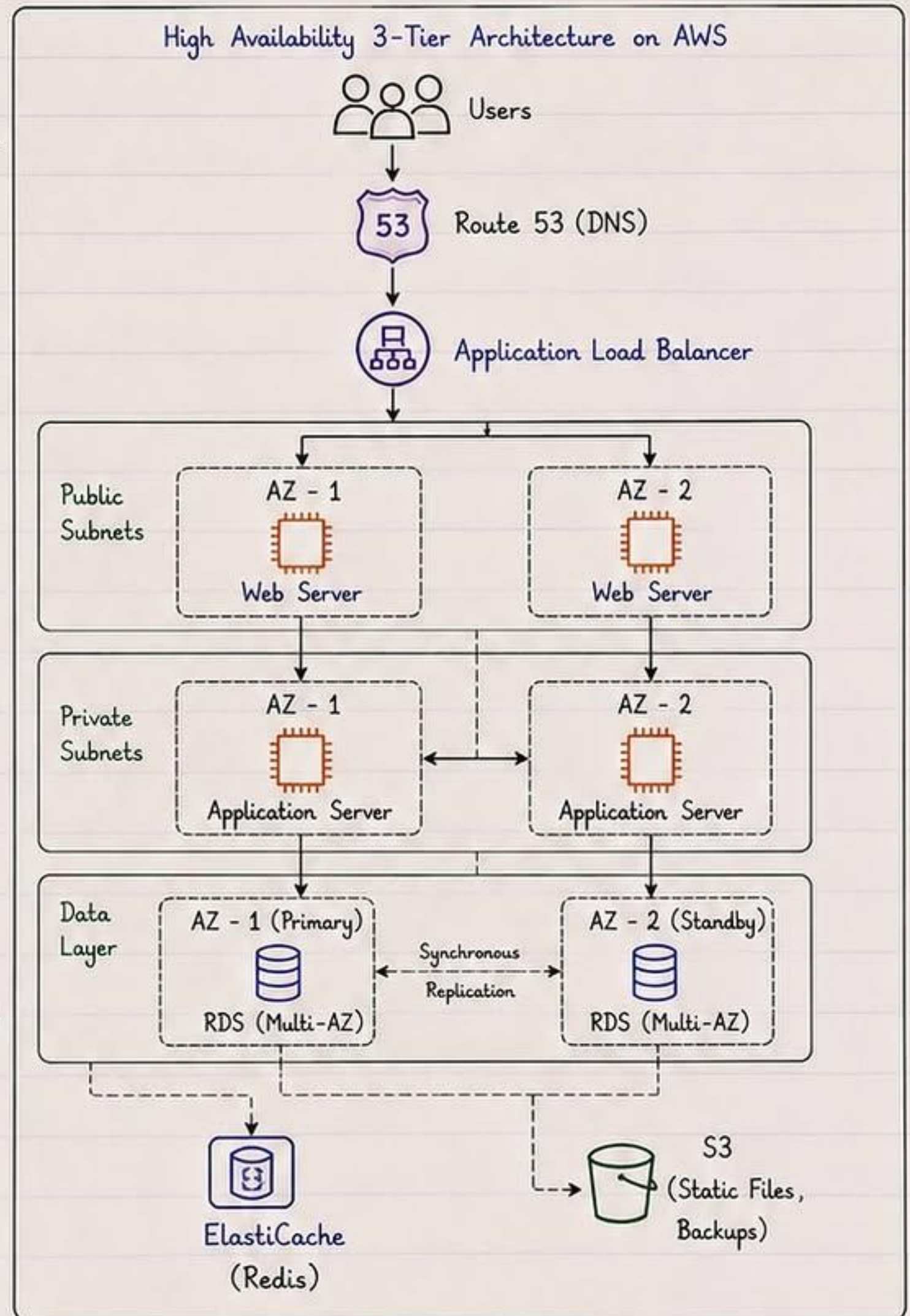
- Distributes incoming traffic across multiple targets.
- Types: ALB (Layer 7), NLB (Layer 4).

5. Caching

- Use Amazon ElastiCache (Redis/Memcached) to reduce database load.
- Improves application performance.

6. Content Delivery Network (CDN)

- Use Amazon CloudFront to deliver content faster to global users.
- Reduces latency and origin load.



Best Practices

- Design for failure and expect it.
- Use managed services wherever possible.
- Enable monitoring and logging.
- Follow least privilege access.
- Automate using Infrastructure as Code.
- Regularly backup and test disaster recovery.

Well-Architected Pillars

- Operational Excellence : Run and monitor systems effectively.
- Security : Protect data and systems.
- Reliability : Ensure availability and fault tolerance.
- Performance Efficiency : Use resources efficiently.
- Cost Optimization : Minimize costs and maximize value.

Interview Question

Q. What is the benefit of Multi-AZ architecture in AWS?

Ans: It provides high availability and fault tolerance. If one Availability Zone fails,

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Architecture Patterns & Best Practices

7. Microservices Architecture

- Break application into small, independent services.
- Each service can be developed, deployed and scaled independently.
- Use Amazon ECS, EKS, Lambda.

8. Serverless Architecture

- Build and run applications without managing servers.
- Use AWS Lambda, API Gateway, DynamoDB, S3 and more.

9. Event-Driven Architecture

- Components communicate via events.
- Use Amazon SNS, SQS, EventBridge, Lambda.

10. Security Best Practices

- Follow the principle of least privilege.
- Encrypt data in transit and at rest.
- Enable MFA and use IAM Roles.
- Regularly audit and monitor resources.

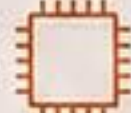
11. Operational Excellence

- Automate repetitive tasks.
- Use Infrastructure as Code (CloudFormation, Terraform).
- Continuously improve processes.

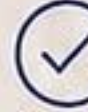
12. Cost Optimization

- Right-size resources.
- Use Reserved Instances and Savings Plans.
- Monitor costs with AWS Cost Explorer.

Common Architecture Patterns on AWS

Web Application	 →  →  → 	For scalable web applications with databases.
Microservices	  	Independent services that communicate via APIs.
Serverless	 →  → 	No servers to manage. Pay for what you use.
Event-Driven	 →  → 	Respond to events and build loosely coupled systems.
Data Lake	 →  → 	Store, process and analyze large amounts of data.

Well-Architected Framework

				
Security	Reliability	Performance Efficiency	Cost Optimization	Operational Excellence
Protect information, systems and assets.	Ensure system availability and manage failure.	Use resources efficiently to meet system requirements.	Avoid unnecessary costs and optimize resources.	Run and monitor systems and improve processes.

Production Tips

- Enable CloudWatch Alarms for critical metrics.
- Use Auto Scaling to handle traffic spikes.
- Store backups in S3 with lifecycle policies.
- Enable AWS Config for compliance tracking.
- Use VPC Endpoints to secure traffic.

Quick Summary

- ✓ Choose the right architecture pattern for your use case.
- ✓ Design for failure, scale and security.
- ✓ Automate deployments and monitoring.
- ✓ Continuously optimize performance and cost.
- ✓ Review and improve your architecture regularly.

Interview Question

Q. Which architecture pattern should you use for an application with unpredictable traffic ?